See My Mind & Noya: Technical Architecture Baseline

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Overview

See My Mind builds on Shuaib's Frappe blueprint (v0.2) and data model (v0.1). Noya, the AI layer, follows the design principles in Malcolm's three AI documents, adapted to that platform.

This replaces an earlier merge of Malcolm's documents. Those documents were written against the original Flask prototype (`app.py`, `noya.py`, `services.py`, SQLite) and proposed an Azure App Service and PostgreSQL stack. The working baseline is now Shuaib's: a mobile-first custom UI over Frappe, with MariaDB.

What stays from Malcolm's work: the AI design rules, child-scoped retrieval, filtering before search, citations, draft-and-approve, safety guardrails and audit logging. **What changes:** hosting, database, sign-in and how access rules are enforced.

Noya's role: an advisory layer only. Facts, observations and AI inference stay separate, with provenance. Noya is not a general chatbot, a substitute for SENCo judgement or a safeguarding decision-maker.

Section 8 lists the questions still open, including those Xinyuan raised in her review.

Platform baseline

Frappe owns the data model, workflow, permissions, audit and backend services; a custom mobile-first See My Mind UI sits on top, and Noya runs as a separate service whose outputs are stored back in Frappe.

Layer	Baseline (Shuaib)	Malcolm's earlier assumption
User experience	Custom See My Mind UI, responsive web/PWA first, Capacitor or native wrapper later	Existing Flask templates
Application	Frappe Framework v16: DocTypes, controllers, roles, hooks, background jobs, reports	Flask app (`app.py`, `services.py`)

Layer	Baseline (Shuaib)	Malcolm's earlier assumption
Database	MariaDB; PostgreSQL not selected for MVP	PostgreSQL Flexible Server
Tenancy	Site/database per school recommended for the prototype (open decision)	Not addressed
Identity	Frappe accounts + mandatory 2FA for staff/admin; school SSO evaluated later	Microsoft Entra ID from the start
Access rules	Role + relationship + visibility + data classification, via Frappe permissions	Reuse of `services.py`
AI/RAG	Separate AI service and vector index; provenance and human review held in Frappe	Azure OpenAI + Azure AI Search
Environments	DEV (test data), UAT (synthetic school data), PROD (real data only after gates)	dev / staging / prod on Azure
Hosting	DEV local Docker; UAT hosted; PROD managed hosting with HTTPS, backups, monitoring	Azure App Service

Mobile apps never bypass Frappe permissions. Browser and mobile call the same permission-aware APIs: dashboard, child profile, create observation, handover, export and AI insight.

Hosting and the AI provider are still open. Malcolm's case for UK/EU-hosted, auditable AI services still applies when choosing them; Azure remains one option for hosting the AI service even with Frappe as the core.

Data model for Noya

The data model has 47 entities; Noya draws on the timeline spine, the support profile and the plans/evidence records, and every Child Event records who reported it and whether it was reported, directly observed, self-reported or system-generated.

Evidence Noya draws on

Entity	What it holds	Build phase
Child Event	Canonical timeline: one row per event, with context, significance, evidence basis, visibility, outcome	MVP (built in POC)
Observation	Wellbeing, learning engagement and outcome, regulation, sensory state, trigger/antecedent, strategy used (free text) and its effectiveness	MVP (built in POC)
Transition Log	Morning handover (sleep, breakfast, medication taken, mood), arrival, lesson change, end-of-day rating	MVP
Child Check-In	Child self-report: mood, feeling, energy, anxiety, help requested	MVP
Behavioural Incident	Antecedent, behaviour, impact, severity, de-escalation, outcome, safeguarding flag	MVP
Strategy Use	Which strategy was used in an event, by whom, effectiveness, child response, outcome, recommend again	Phase 2
Child Support Profile	Communication style, support, sensory, distress and calming summaries, non-negotiables (mostly free text)	MVP
Child Trigger / Distress Indicator	Structured triggers (type, context, severity, mitigation) and early signs of distress	MVP
Child Strength	Structured strengths with evidence and confidence (not diagnostic)	Phase 2
Support Strategy / Child Support Strategy	Strategy catalogue (name, category, description) and child-specific use, priority, baseline effectiveness	MVP

Entity	What it holds	Build phase
Child Plan / Plan Outcome / EHCP Review	Versioned plans, targets with success criteria and status, review meetings	MVP
Communication Thread / Message	Parent-school messages that form part of the child record	MVP
Child Document	EHCPs, reports, letters, assessments; carries visibility and an RAG-eligible flag	MVP
Learning Worksheet / Assignment / Progress	Worksheets, adaptations, completion and scores	Future (bundled with Parent Plus)

How the ABC chain is captured

- **Antecedent:** parent morning handover (Transition Log), plus the antecedent fields on Observation and Behavioural Incident.
- **Intervention:** Strategy Use (Phase 2); until then, free text in Observation.
- **Outcome:** Child Check-In, the outcome fields on events and Strategy Use, and Plan Outcome progress.

Records are tied together by child and date through the Child Event timeline, not by explicit event-to-event links.

AI records

AI Run (requester, model, prompt version, status), AI Insight (insight type, confidence, human review status, reviewer) and AI Insight Evidence (citations to a Child Event or document chunk) keep AI output separate from fact. RAG Source Document and RAG Chunk Index track indexed documents. All are Phase 2.

Noya AI design principles

Both Malcolm's and Shuaib's documents agree on how Noya must behave; these rules are enforced in retrieval and the data model, not by prompt instruction alone.

1. **Child-scoped access only.** Every query covers one child and only what the requesting user may see. Parents see only their own child or children.
2. **Human review before sharing.** AI output is a draft. Staff review it before it is shared outside the agreed audience (the review tiers are an open question, section 8).

3. **Evidence-first.** Every claim cites the event or document it came from, by date and record type. Thin evidence is stated as such, never filled in.
4. **SEND guardrails.** No diagnosis, clinical opinion, safeguarding decision or autonomous action. Patterns are flagged for a human to assess.
5. **Facts, observations and inference stay separate.** AI output is stored as an AI Insight with its evidence, model, confidence and review status, and is never the record of what happened.
6. **Data minimisation.** Only the data needed for the specific AI task is sent to the model.
7. **Full audit.** Every AI run logs who asked, which child, what was retrieved and what was produced.
8. **Separate purposes need separate consent.** Parent Plus, careers and research reuse are opt-in, with their own consent.

Planned Noya features

Daily teacher briefing · SENCo trend summary · parent-friendly daily digest · support strategy suggestions · meeting note summaries · pattern alerts · EHCP and review summary drafting.

Retrieval and generation design

Malcolm's retrieval design carries over with two adaptations: access rules come from Frappe permissions instead of `services.py`, and timeline events are indexed alongside documents, not only uploaded files.

What gets indexed

- **Timeline events:** Child Event with its detail record (observation, transition, check-in, incident, strategy use), one chunk per event. The current data model only defines RAG records for documents; AI Insight Evidence can already cite a Child Event, so this is an extension rather than a redesign.
- **Documents:** Child Documents flagged RAG-eligible (EHCPs, reports, assessments), OCR'd and chunked by section or page.
- **Profile context:** the Child Support Profile, triggers and active strategies, always included as static context.
- **Earlier Noya output:** reviewed AI Insights, as derived summaries.

Chunk by event or document, never by fixed token windows, so each unit stays dated and citable.

Metadata on every chunk

Child, school, date, record type, reporter type and evidence basis, visibility, data classification and consent status. Malcolm's sensitivity levels (standard, elevated, restricted) map onto Shuaib's classifications (personal, sensitive SEND, safeguarding/high risk, AI generated); the two vocabularies need to be merged into one.

Two-stage retrieval

1. **Hard filter first:** this child, only records the user's role, relationship and visibility allow, consent not withheld. Run as an index filter, so the model never sees excluded records.
2. **Search second:** vector plus keyword search within the filtered set, ranked by relevance and recency, returning roughly 6–12 chunks.

A sync job indexes new and changed records from MariaDB incrementally (no duplicates), and logs every run.

Generation

Every call uses this instruction pattern:

> Answer using only the evidence provided below. Every claim must cite the record(s) it came from by date and source type. If the evidence is thin, insufficient, or contradictory, say so explicitly rather than inferring. Do not offer a diagnosis, clinical opinion, or safeguarding conclusion — flag patterns for human review only.

- Scope (child, date range, record types) is passed explicitly; never an open "search everything" instruction.
- "Limited evidence found" is required below a minimum number of relevant records (for example, fewer than 3 for a trend).
- Output is structured (briefing, answer, suggested next step) and saved as an AI Insight with its evidence links.
- Free-text notes and messages are checked for instruction-like text before they reach the model.

Privacy, security and go-live gates

No real child or SEND data enters the system until privacy, security, tenancy, data, operations and UAT gates are met with evidence.

Privacy model

- Every field is classified: personal, sensitive SEND, safeguarding/high risk, system metadata or AI generated.
- Purpose and lawful basis are attached to capture and reuse (support, safeguarding, communication, reporting, Parent Plus, research).
- Provenance is recorded: who reported, observed, generated and reviewed each record.
- Consent is per child and guardian relationship and per purpose, with version and withdrawal history (Consent Record).
- Retention, deletion, archive and export are designed before real data enters production.

Access and security

- Access is derived from role + relationship + record visibility + data classification.
- Access is tested negatively: unassigned staff cannot open a child, a guardian cannot see a non-linked child, School A cannot see School B, restricted notes stay restricted, exports respect visibility.
- Mandatory 2FA for staff and admin; guardian 2FA considered against usability.
- Every view, edit, export, permission change and support access to sensitive records is audited; support staff get break-glass access with reason and approval.
- Behavioural incidents carry a safeguarding flag that routes to the school's own safeguarding process, never replacing it.

Go-live gates for real data

Gate	Evidence required
Privacy	DPIA, lawful basis, consent model, retention/deletion, privacy notices
Security	2FA, HTTPS, RBAC, record access tests, pen test, vulnerability remediation
Tenancy	Cross-school access tests, support access controls, backup isolation
Data	Import templates, export packs, data dictionary, data quality checks
Operations	Monitoring, backups, restore test, incident process, support model

Gate	Evidence required
UAT	Parent, teacher, TA, SENCo and admin scenarios passed with synthetic data

Build roadmap

Build a thin but complete vertical slice first; Noya comes last, once the data controls have matured and a term of structured data exists.

MVP sequence (Shuaib)

1. **Foundation:** school, child, guardian, staff, assignments, entitlement basics.
2. **Timeline:** Child Event backbone, observation detail, profile page.
3. **Daily flow:** morning handover, teacher observation, end-of-day summary.
4. **Workflow:** notifications, concerns, follow-up actions, SENCo review trigger.
5. **Dashboards:** parent, teacher, TA and SENCo views using real data.
6. **Assurance:** UAT, privacy controls, pen test, export/import, backup/restore.
7. **Mobile packaging:** PWA first, then app stores once flows are stable.
8. **AI/RAG pilot:** advisory summaries and evidence retrieval.

Noya phases within step 8 (from Malcolm's RAG design)

1. **Structured answers:** scoped questions answered straight from Frappe records with templated generation; no vector index. Tests citations and review cheaply.
2. **Retrieval layer:** index timeline events and add two-stage retrieval.
3. **Documents:** EHCPs, reports and scans via OCR, chunked by section.
4. **Workflow integration:** meeting notes, parent digests, intervention summaries, SENCo pattern review, all draft-and-approve.

Recommended changes to the data model before build

- **Move Strategy Use into the MVP.** Otherwise the pilot term captures strategies only as free text, and the strategy→outcome link Noya most needs is lost.
- **Add a structured needs record** alongside the free-text support profile; the blueprint names a Support Need entity that is not yet in the workbook.
- **Add the six UCL pilot categorisation fields** (EHCP status, primary diagnosis, medication Y/N, communication profile, support level, school type), with diagnosis held as context, never a retrieval key.

- **Separate classroom learning evidence from Parent Plus,** so worksheets and work samples can be captured in the pilot.
- **Extend RAG records to timeline events,** not only documents.
- **Merge the sensitivity vocabularies** used by Malcolm and Shuaib into one classification.

Open design questions

The architecture covers safety, access and infrastructure well; what is still undefined is what Noya knows, how it reasons, and how anyone knows it is right. Owners are suggested.

#	Question	Where it stands	Suggested owner
1	How are a child's SEND needs represented for retrieval, without categorising children by diagnosis? (Xinyuan)	Structured triggers, distress indicators and strengths exist; support profile is mostly free text; no Support Need entity yet	Xinyuan + Shuaib
2	How are events, strategies, outcomes and learning evidence linked so insights rest on the child's own evidence, not the model's general knowledge? (Xinyuan)	Timeline and Strategy Use designed; Strategy Use is Phase 2; worksheets are Future; RAG records cover documents only	Xinyuan + Shuaib
3	How does Noya detect patterns: rules, statistics or trained models, and at what thresholds?	One rule defined (N concerns in N days); method otherwise undefined	Xinyuan
4	What evidence base sits behind the strategy catalogue, and which needs does each strategy suit?	Catalogue holds name, category and description only	Jonathan + Xinyuan

#	Question	Where it stands	Suggested owner
5	What exactly is ruled out by "no labelling"? Proposed tiers: diagnosis or classification (never), patterns and warning signs with evidence (allowed), strategy suggestions (allowed, as drafts). (Xinyuan)	Not defined	Jonathan + Xinyuan
6	What needs human review, by whom and at what stage, at the scale of a whole class? (Xinyuan)	AI Insight has a review status and reviewer; no tiers by audience or sensitivity	Jonathan + Xinyuan
7	What counts as "external sharing": the wider school team, parents, the child's SEND specialist, other agencies? (Xinyuan)	Undefined; External Professional role is Phase 2	Jonathan
8	Where do safeguarding flags and pattern alerts go: SENCo, DSL, or both?	Incidents route to the school's safeguarding process; DSL role not modelled	Jonathan
9	How will we know Noya's insights are accurate and useful?	No evaluation plan; accepted/edited/rejected review data could be the core measure	Xinyuan
10	Who records consent, at what level, and what happens on withdrawal?	Consent Record designed per purpose; process not defined	Jonathan + Shuaib

Platform decisions still open (from Shuaib's blueprint)

- Tenant model: site per school, shared database or hybrid.
- Mobile approach: PWA first, Capacitor wrapper or native.
- Identity: Frappe accounts first; when to add school SSO.
- AI provider and RAG architecture: vector store, model gateway, retention, data minimisation, UK/EU hosting.
- Data controller/processor model and school contract responsibilities.

Sources

- *See My Mind Technical Blueprint v0.2*, Shuaib Rabbani, 15 September 2026
- *See My Mind Logical Data Model & Database Schema v0.1* (workbook), Shuaib Rabbani
- *NOYA Azure Cloud Setup Recommendation*, 20 September 2026, shared by Malcolm Wiredu in #all-seemymind
- *NOYA Future AI Architecture*, 20 September 2026, shared by Malcolm Wiredu
- *NOYA RAG Design Document*, 20 September 2026, shared by Malcolm Wiredu
- Xinyuan Hao, email review of the technical blueprint, October 2026